A ML ALS Survival Model Lacking Vital Capacity for use in Clinical Trials during the COVID-19 Pandemic

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Introduction

The ongoing COVID-19 pandemic has made it difficult in some clinical settings to measure vital capacity for the purpose of eligibility determination. We developed a machine learning survival model without the use of baseline vital capacity measures and asked whether it could stratify clinical trial patients and a wider ALS patient population derived from a tertiary care ALS center (Emory University).

Methods

A machine-learning gradient boosting machine survival model (noVCSurv) lacking baseline vital capacity measures was trained using the PRO-ACT database and compared to a previously validated survival model that included vital capacity (VCSurv). Kaplan-Meier survival curves using strata generated by the noVCSurv model were compared to strata created using baseline vital capacity cut-offs for a clinical trial data set and a clinic data set.

Variable Importances

Feature	Rel. imp. (%) w/ VC	Rel. imp. (%) w/o VC	Chg. in rel. imp. (%)	Rank w/ VC	Rank w/o VC	Chg. in rank
Vital capacity slope (symptom onset to BL)	14.99	NA	NA-	1	NA	NA-
Age (yrs)	12.42	13.40	0.98 ↑	2	1	1 ↑
BL percent expected vital capacity (%)	11.74	NA	NA-	3	NA	NA-
BL BMI	5.27	7.59	2.32 ↑	4	2	2 ↑
Bulbar subscore slope (symptom onset to BL)	4.10	7.02	2.93 ↑	6	3	3 ↑
ALSFRS-R slope (symptom onset to BL)	2.96	5.31	2.35 ↑	10	4	6 ↑
BL pulse (bpm)	3.09	5.25	2.16 ↑	8	5	3 ↑
BL vital capacity (L)	5.10	NA	NA-	5	NA	NA-
Turning in bed (Q7) score slope (symptom onset to BL)	2.68	3.90	1.22 ↑	12	6	6 ↑
Speech (Q1) score slope (symptom onset to BL)	NA	3.87	NA+	NA	7	NA+
(Q6 + Q7) subscore slope (symptom onset to BL)	2.95	3.78	0.83 ↑	11	8	3 ↑
Gross motor subscore slope (symptom onset to BL)	3.01	3.77	0.76 ↑	9	9	0 =
BL weight (kg)	2.46	3.65	1.19 ↑	14	10	4 ↑
Fine motor subscore slope (symptom onset to BL)	3.39	3.62	0.23 ↑	7	11	-4 ↓
BL expected vital capcity	2.52	3.52	0.99 ↑	13	12	1 ↑
Dressing and hygiene (Q6) score slope (symptom onset to BL)	2.42	2.97	0.55 ↑	16	13	3 ↑
Climbing stairs (Q9) score slope (symptom onset to BL)	2.31	2.77	0.46 ↑	17	14	3 ↑
Handwriting (Q4) score slope (symptom onset to BL)	NA	2.52	NA+	NA	15	NA+
Time since disease diagnosis (days)	NA	2.50	NA+	NA	16	NA+
(Q4 + Q5) subscore slope (symptom onset to BL)	2.44	2.32	-0.12 ↓	15	18	-3 ↓
BL ALSFRS-R total score	2.08	2.36	0.28 ↑	18	17	1 ↑
Walking (Q8) score slope (symptom onset to BL)	NA	2.10	NA+	NA	19	NA+
Time since symptom onset (days)	1.87	2.09	0.22 ↑	21	20	1 ↑
(Q8 + Q9) subscore slope (symptom onset to BL)	2.07	1.70	-0.38 ↓	19	25	-6 ↓
BL systolic blood pressure (mmHg)	2.04	1.77	-0.27 ↓	20	22	-2 ↓
BL bulbar subscore	0.65	1.94	1.29 ↑	27	21	6 ↑
Height (cm)	1.86	1.60	-0.26 ↓	22	27	-5 ↓
Cutting (Q5) score slope (symptom onset to BL)	NA	1.75	NA+	NA	23	NA+
BL gross motor subscore	1.33	1.70	0.37 ↑	24	24	0 =
BL (Q8 + Q9) subscore	1.66	1.67	0.01 ↑	23	26	-3 ↓
BL climbing stairs (Q9) score	1.21	1.60	0.38 ↑	25	28	-3 ↓
BL (Q6 + Q7) subscore	NA	1.07	NA+	NA	29	NA+
BL speech (Q1) score	0.53	0.87	0.33 ↑	28	30	-2 ↓
Sex	0.66	NA	NA-	26	NA	NA-
BL turning in bed (Q7) score	0.18	NA	NA-	29	NA	NA-
Flag: ALS symptom onset (bulbar)	NA	0.03	NA+	NA	31	NA+

In the table above, a value of NA represents a variable that has either been added to (+) or removed from (-) the model trained w/o vital capacity features. Changes in relative importance and rank are green to represent an increase from the original model, red to represent a decrease, and blue to represent no change.

Model with vital capacity:
GBM, 30 variables, AUC=0.815
n=5,830 including 1,750 deaths

Model without vital capacity:
GBM, 31 variables, AUC = 0.781
n=5,828, including 1,748 deaths

noVCSurv Model Performance

Model	Validation dataset (type)	AUC	Discrimination (C-index)	Calibration (log rank test failures)
VCSurv	PRO-ACT (internal)	0.815	0.196	10/30
	VITALITY-ALS (external clinical trial)	0.777	0.211	0/3
	Emory (external clinic)	0.779	0.224	3/3
noVCSurv	PRO-ACT (internal)	0.781	0.232	4/30
	VITALITY-ALS (external clinical trial)	0.748	0.239	1/3
	Emory (external clinic)	0.792	0.209	3/3

AUC = area under the ROC curve

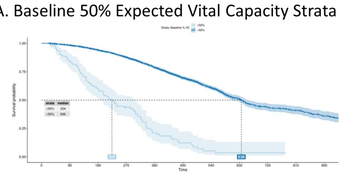
The survival model degrades a bit with the removal of baseline vital capacity, but remains usable for discrimination (i.e., stratification).

Results

The noVCSurv model suffered a minor decline in performance compared to the VCSurv model. When analyzed using clinical trial data, the noVCSurv model replicated the strata created using baseline vital capacity. When analyzed using the Emory Clinic data, the noVCSurv model outperformed strata created using baseline vital capacity.

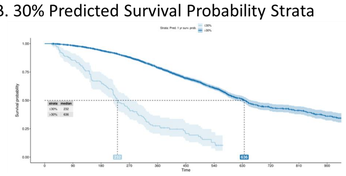
PRO-ACT: 50% Baseline Vital Capacity Strata vs. 30% Predicted Survival Probability Strata

A. Baseline 50% Expected Vital Capacity Strata



- Median survival:
  - Excluded: 224 days
  - Included: 636 days
- Exclude 2.73% of the PRO-ACT population

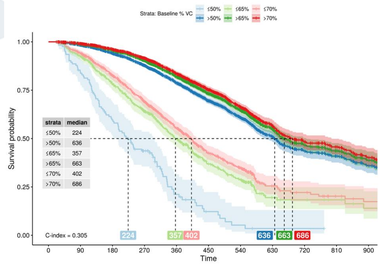
B. 30% Predicted Survival Probability Strata



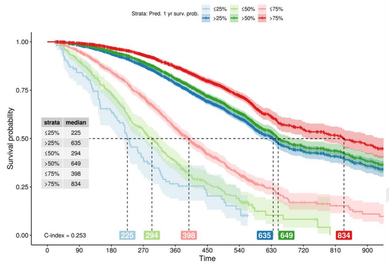
- Median survival:
  - Excluded: 232 days
  - Included: 636 days
- Exclude 3.20% of the PRO-ACT population

PRO-ACT: Comparison of Baseline Vital Capacity Strata vs. Predicted noVCSurv Survival Model Strata

A. Baseline Vital Capacity Strata



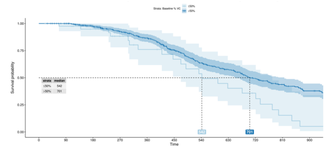
B. Strata Created using "VC-Free" Survival Model



Data used for the development of the models in this study were obtained from the PRO-ACT Database. Data used in testing the models were obtained from the Emory University Clinic courtesy of Dr. Jon Glass and from Cytokinetics, Inc.

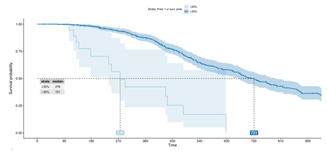
Clinic Data: 50% Baseline Vital Capacity Strata vs. 30% Predicted Survival Probability Strata

A. Baseline 50% Expected Vital Capacity Strata



- Median survival:
  - Excluded: 542 days
  - Included: 701 days
- Exclude 9.3% of the Emory population

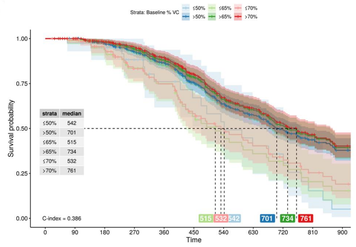
B. 30% Predicted Survival Probability Strata



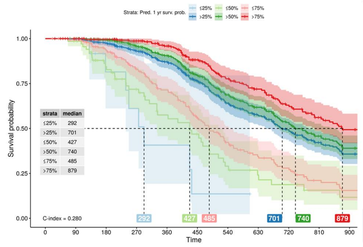
- Median survival:
  - Excluded: 276 days
  - Included: 721 days
- Exclude 4.2% of the Emory population

Clinic Data: Comparison of Baseline Vital Capacity Strata vs. Predicted noVCSurv Survival Model Strata

A. Baseline Vital Capacity Strata



B. Strata Created using "VC-Free" Survival Model



Conclusions

- A survival model that doesn't use baseline vital capacity as a predictor can be built using machine learning
  - Model degrades little compared to model that includes VC
  - Model remains useful for stratification
- The "VC-free" survival model returns useful predictions for a clinical trial population as well as a broad ALS clinic population
- The model can be used to stratify ALS patients for ALS clinical trials in the absence of baseline vital capacity measurements
  - Useful for inclusion/exclusion during COVID19 pandemic